

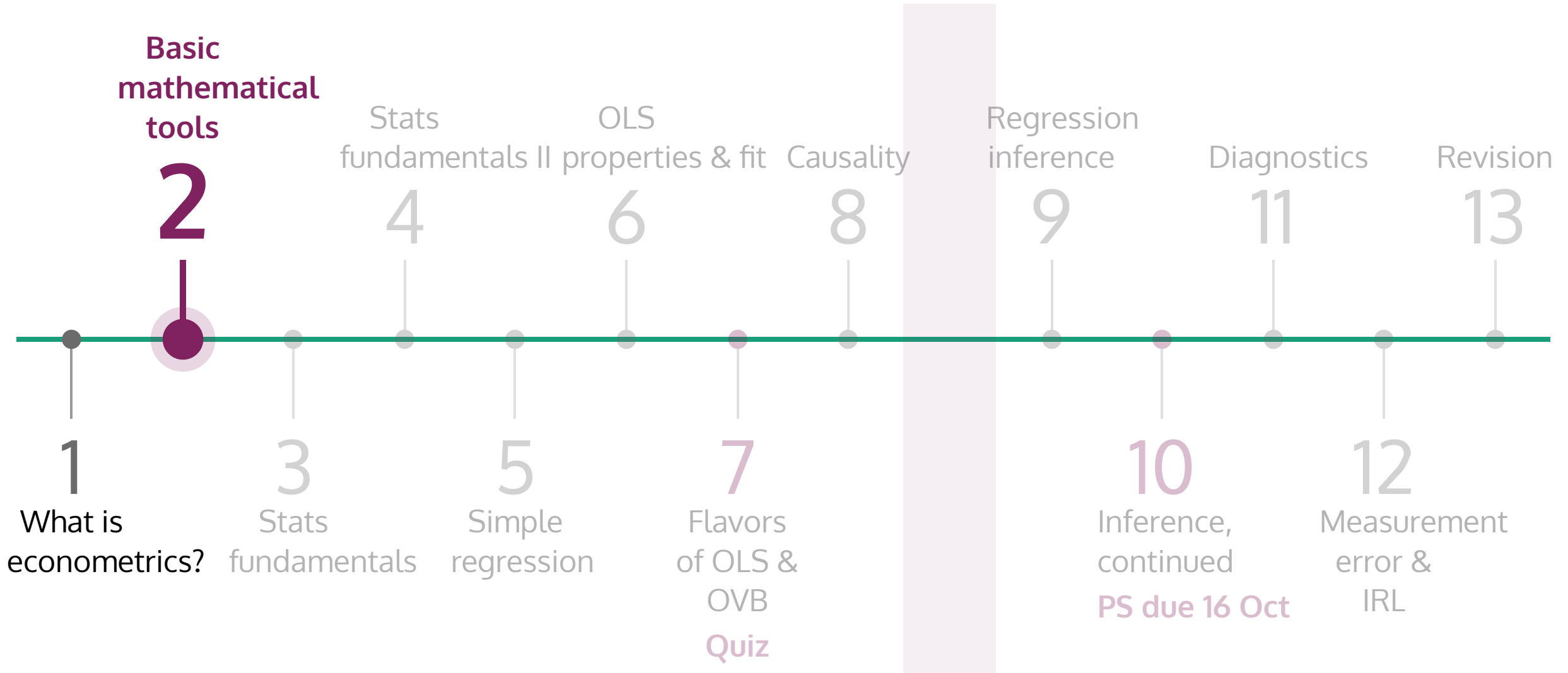
A vibrant explosion of pink, orange, and blue powder against a black background. The powder is captured in mid-air, creating a starburst effect with various shades of magenta, fuchsia, and cyan. The explosion is centered in the upper half of the frame, with the powder trailing off towards the bottom and sides.

ECON 2041

Introductory Econometrics

Get using Python

Plan for the semester



Plan for today

- Why you should learn to code
- What is Python & why are we using it?
- Setting up Google Colab
- Python basics

The elephant in the (class)room

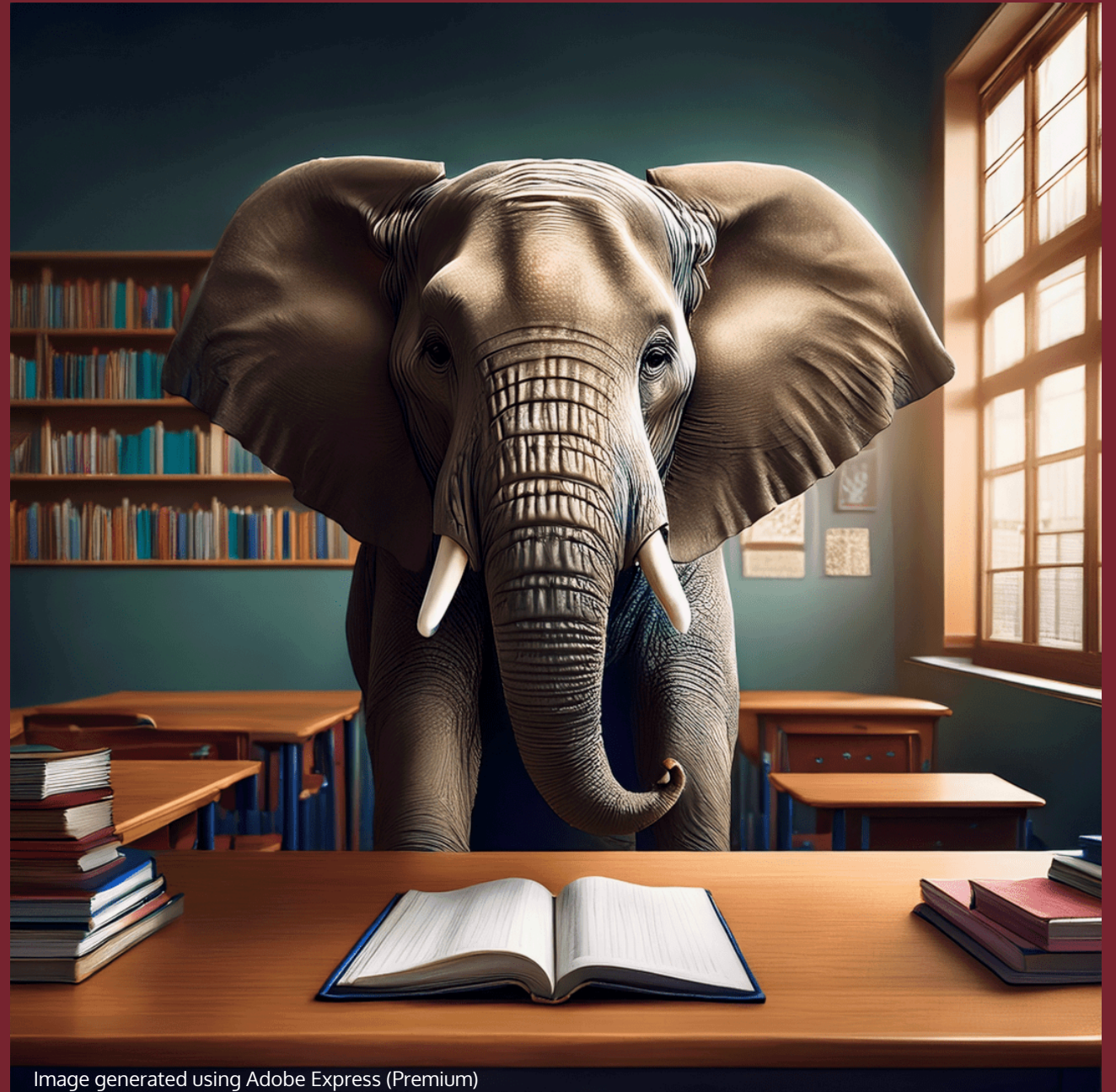
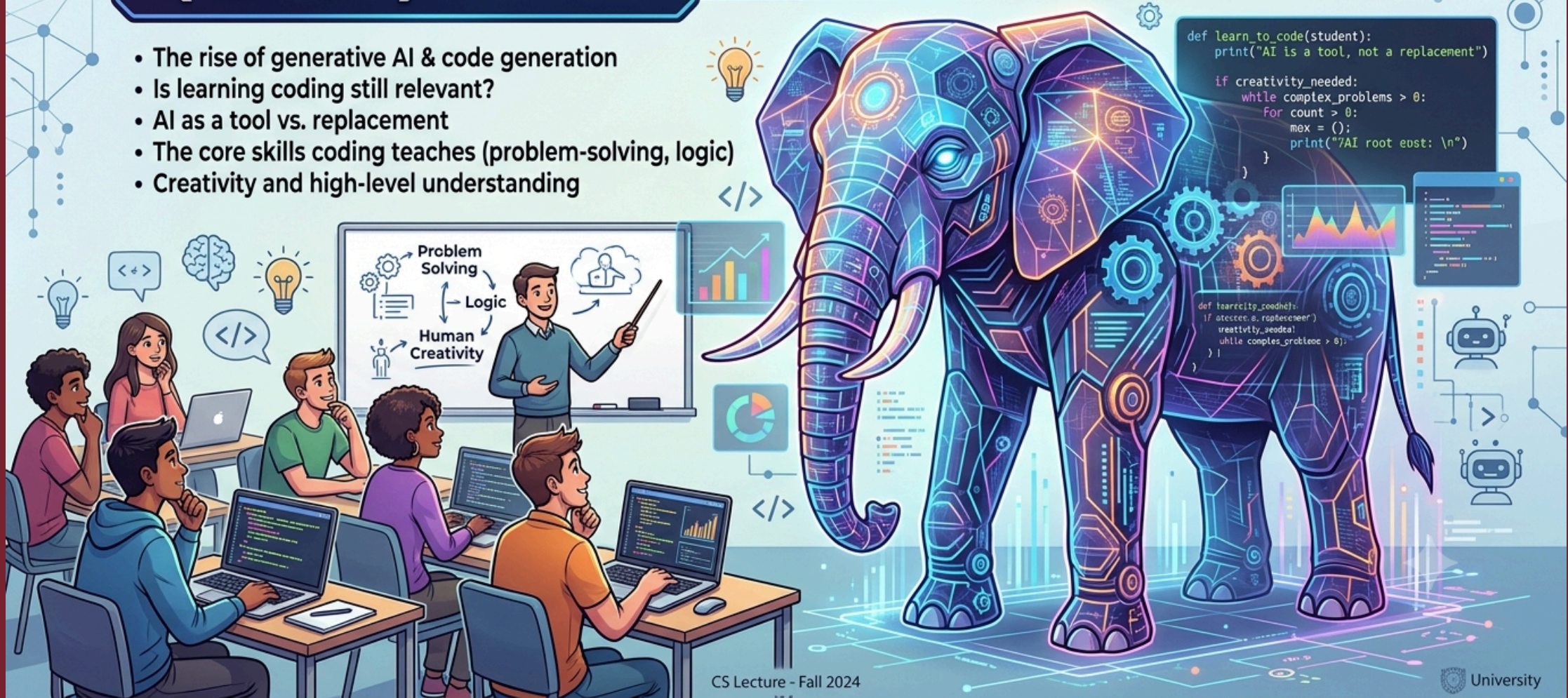


Image generated using Adobe Express (Premium)
Prompt: elephant in the classroom

'ELEPHANT IN THE (CLASS)ROOM'

ABOUT AI AND WHY STUDENTS SHOULD STILL LEARN TO CODE

- The rise of generative AI & code generation
- Is learning coding still relevant?
- AI as a tool vs. replacement
- The core skills coding teaches (problem-solving, logic)
- Creativity and high-level understanding



The Effects of Climate Change

- Global temperature increase
 - sea level rise
- More extreme weather events
- Loss of biodiversity and habitat
- Impacts on agriculture and food security
- Human health consequences
- Migration and displacement

How can we mitigate these effects?

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- Loss of biodiversity and habitat
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How can we mitigate these effects?



What's your background?

Quick survey on your background & interests

1. Open your portal link
<https://emiliatjernstrom.com/econ2041/portal.html>

OR scan the QR code, then enter your code

2. Answer the five questions and hit Submit

Portal answers are anonymous & ungraded



Why you should still learn to code

... even though AI can (sort of) do it for you

Coding builds core **transferable** skills

- teaches problem solving
- sharpens logic & structure
- trains you to check & debug

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AI can't replace understanding

- AI guesses but you still need to judge
- bad code may still run
- without knowing the basics you can't fix errors

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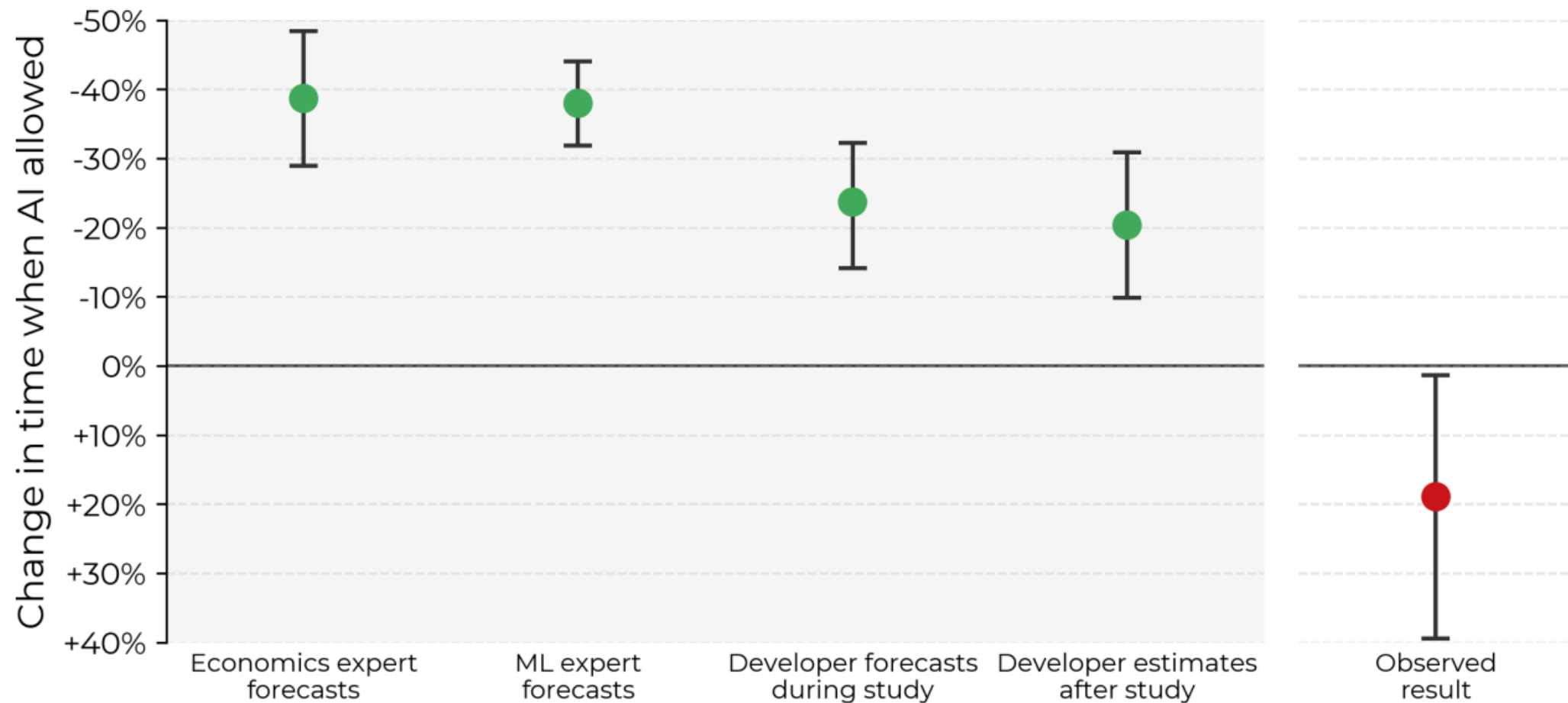
Coding makes you stand out

- signals technical fluency
- shows initiative & curiosity
- you will do more than peers who rely on AI

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Experts and Study Participants Misjudge AI Speedup



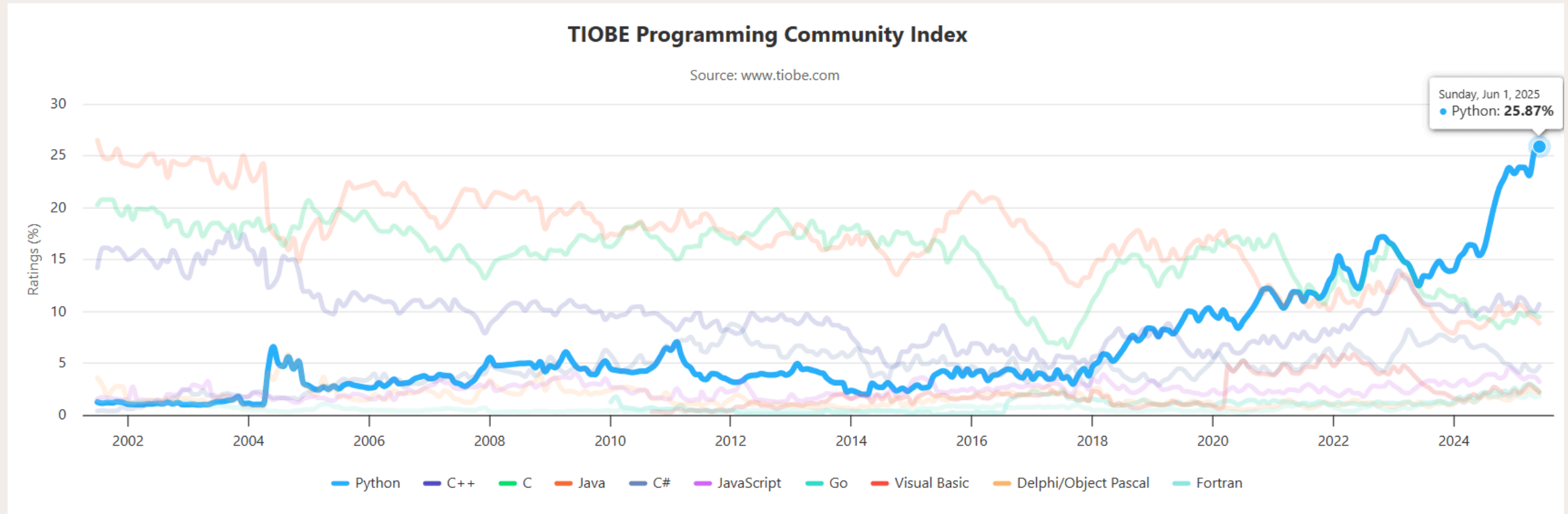
What is Python?

Python!

We'll use Python & Jupyter notebooks for our econometric data work

Python is...

one of the most popular programming languages



Python!

We'll use Python & Jupyter notebooks for our econometric data work

Python is... a free & open-source general-purpose programming language

used **for**

- AI
- scientific computing
- communication
- web development
- CGI and graphical user interfaces
- game development
- etc.

Python!

We'll use Python & Jupyter notebooks for our econometric data work

Python is... a free & open-source general-purpose programming language

used **by**

- Google
- OpenAI
- Netflix
- Meta
- Dropbox
- Amazon
- Reddit
- etc.

Jupyter notebooks

We'll use Python & Jupyter notebooks for our econometric data work

Jupyter notebooks: one of the many ways of interacting with Python

They let you use your browser to

- write & execute Python commands
- format output, including tables, figures, animations, etc.
- combine text, mathematical expressions, and code

Jupyter notebooks

We'll use Python & Jupyter notebooks for our econometric data work

Jupyter notebooks: one of the many ways of interacting with Python

Jupyter isn't the only way to code in Python, but it's GREAT for

- getting started coding in Python
- testing new ideas
- interacting with small pieces of code
- using online interactive environments such as Google Colab
- sharing / collaborating on research

We work in Colab

Everything in this unit runs in **Google Colab**,
in your browser

- nothing to install
- works on any machine you sit at
- your work saves to your Google Drive



Google Colab

Colab allows you to run Python (in a Jupyter notebook) on a Google-provided remote cloud computer

Amazingly, this is free but it does require a Google account

One cool feature of this is that you can run it from your smartphone

Over to the notebook

Open this week's lecture notebook and follow along:

<https://emiliatjernstrom.com/econ2041/lectures/w02>

It will open directly in Colab

Click **Copy to Drive** first, so your edits save

Now you know...

- How to open and run a Jupyter notebook
- How to get Python to print something for you
- How to use Python as a calculator

I hope you're excited about applying these skills and building on them!